

3. Rotation Matrix (II): Diagonalization of Quadratic Form

In general, a conic can be described by the quadratic equation

$$\begin{bmatrix} x & y \end{bmatrix} M \begin{bmatrix} x \\ y \end{bmatrix} = 1. \quad (1)$$

where the matrix M reads

$$M = \begin{bmatrix} A & B/2 \\ B/2 & C \end{bmatrix}.$$

For example, if $A = a^{-2}$, $C = b^{-2}$ and $B = 0$, Eq. (1) will reduce to the standard form of an ellipse $a^{-2}x^2 + b^{-2}y^2 = 1$ with semi-major axis a and semi-minor axis b . It is just because of $B = 0$ (i.e. the matrix M in Eq. (1) is *diagonal*) that we are lucky to have the major and minor axes falling on the x and y axes respectively. However it is generally not the case.

When $B \neq 0$, the major/minor-axis of the ellipse will be rotated away of the axes of the coordinate system. To see how the ellipse is rotated, let us consider a rotation matrix $R(\theta)$ that rotates the vector $\begin{bmatrix} x & y \end{bmatrix}^T$ (counterclockwise) by the angle θ to $\begin{bmatrix} x' & y' \end{bmatrix}^T$,

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = R(\theta) \begin{bmatrix} x \\ y \end{bmatrix},$$

under which the conic equation becomes

$$\begin{bmatrix} x' & y' \end{bmatrix} M' \begin{bmatrix} x' \\ y' \end{bmatrix} = 1,$$

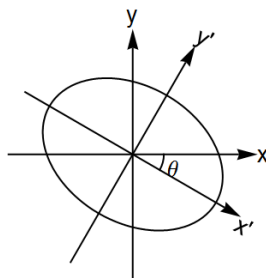
where

$$M' = \begin{bmatrix} A' & B'/2 \\ B'/2 & C' \end{bmatrix}.$$

(a) Work out A' , B' and C' in terms of A , B , C and θ .

(b) Show that $\text{Tr } M = \text{Tr } M'$ and $\det M = \det M'$, i.e. the trace and determinant of the matrix M is invariant under rotation.

The transform of coordinates is effectively the same as the transformation of the reference frame in the *reverse* manner. If in the x' - y' frame, the major/minor axis falls on the axes of the frame, we say that the quadratic form is diagonalized in the x' - y' frame.



(c) What θ should be to diagonalize the quadratic form in the x' - y' frame?

When the quadratic form is diagonalized, the semi-major/minor axis can be determined from the diagonal elements.

(d) Find the semi-major axis a and semi-minor axis b of a conic described by Eq. (1) in terms of $\text{Tr } M$ and $\det M$.

(e) Show that the conic will be an ellipse (or circle) if $\det M > 0$.

Note: a 2×2 real symmetric matrix M can always be diagonalized by a $SO(2)$ transform $R(\theta)$ (i.e. a rotation in the 2-dimensional space). The geometric meaning of diagonalization

is to rotate the conic so that its major/minor axis coincides with the frame axes. After diagonalization, the diagonal elements are known as the eigen values of the original matrix, which are invariant under rotation, providing information of the shape of the conic.

Chap. 13 Work and Potential Energy (A)

1. Gravitational Pull from Interstellar Dust

Consider a planet of mass m is orbiting around the Sun (whose mass is M) on a circle of radius r . Assume that there is uniform interstellar dust of the density ρ (mass per volume) in the solar system.

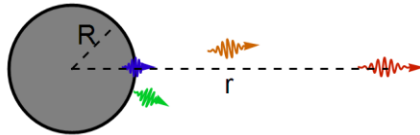
(a) Show that the planet will experience an extra gravitational force provided by the dust of the magnitude $F' = -mKr$, where $K = (4\pi/3)\rho G$. (Neglect the resistance due to collisions with the dust.)

(b) If the angular momentum of the planet is L , what is the radius r of the circular orbit in terms of L , G , M and ρ ?

2. Gravitational Red Shift (Newtonian Limit)

Gravitational red shift refers to the phenomena that light emitted from stars undergoes frequency shift to red-end due to gravitation. According to quantum mechanics, the energy of a photon is $E = h\nu$, where h is the Planck constant and ν the photon frequency.

(a) What is the inertial (and gravitational) mass of photon?



(b) Given the photon frequency ν and the mass of the star M , what is the potential energy of photon in the gravitational field of the star (according to Newton's Law of Gravity), if the photon is r distant away from the center of the star?

(c) Suppose the photon travels from r to $r + dr$, and its frequency changes from ν to $\nu + d\nu$. What is the corresponding change in the potential energy?

The increase of the potential energy as the photon climbs out of the potential well of the star should be compensated by the decrease of its kinetic energy such that the total energy is conserved. Since the kinetic energy of the photon is proportional to its frequency, this will leads to the drop of photon frequency as it flies away from the star, and hence the gravitational red shift.

(d) According to the conservation of energy, deduce the photon frequency ν as a function of r , given that the frequency when the photon is emitted from the surface of the star ($r = R$) is $\nu(R)$.

(e) Find the red shift $\Delta\nu/\nu$ of light of a far star given its mass M and radius R , where $\Delta\nu = \nu(\infty) - \nu(R)$.

(f) It is observed^[1] that the light from the white dwarf Sirius B suffers the gravitational red shift $\Delta\nu/\nu = 2.7 \times 10^{-4}$. The mass of Sirius B is roughly the same as the solar mass, estimate the radius and hence the mass density of Sirius B from its gravitational red shift.

Note: The calculation of gravitational red shift in this problem is based on Newton's Law of Gravity, which does not take into account the effect of general relativity. The general theory of relativity is known as a more correct theory of gravity, especially when the gravitational field is strong or fast varying. To estimate the influence of general relativity, we introduce the Schwarzschild radius $R_S = 2GM/c^2$. A star will become a black hole if its radius is less than its Schwarzschild radius, which means that light cannot escape from the surface of the star. From the viewpoint of gravitational red shift, this means light emitted from a black hole will red shift infinitely, i.e. its frequency drops to zero before travelling far. In this case, general relativity is important to calculate the red shift correctly. The precise result given by general relativity is $\nu(\infty) = \nu(R)(1 - R_S/R)^{1/2}$. Our Newtonian approximation is valid only in the limit when $R \gg R_S$, say the radius of the star is far greater than the radius of a black hole of the same mass.

Reference

[1] M.A. Barstow, H. E. Bond, J.B. Holberg, M.R. Burleigh, I. Hubeny and D. Koester, Mon. Not. R. Astron. Soc. **000**, 1-11 (25-May-2005).

Chap. 14 Work and Potential Energy (B)

1. Potentials of Conservative Forces

Calculate the potential of the following forces. I should also tell you what is the conventional choice of the potential zero point.

- (a) Gravity, near the Earth's surface: $F = -mg$, potential zero point: $V(z = 0) = 0$.
- (b) Gravity, between masses: $F = -Gm_1m_2/r^2$, potential zero point: $V(r = \infty) = 0$.
- (c) Electrostatic force, between charges: $F = q_1q_2/(4\pi\epsilon_0r^2)$, potential zero point: $V(r = \infty) = 0$.
- (d) Ideal spring: $F = -kx$, potential zero point: $V(x = 0) = 0$.
- (e) Nuclear force: $F = -(1/r^2 + 1/(rr_0)) \exp(-r/r_0)$, potential zero point: $V(r = \infty) = 0$.

2. The Choo-choog Machine (III): Work and Energy

Let us go back to the choo-choog machine once more. A weight of mass M is held up by the elbow like pivoted rods of the lengths l . The roller is being driven by some machinery at a constant speed v .



When the distance between the roller and the pivot is l ,

- (a) What is the energy E (both kinetic and potential) of the weight?
- (b) What is the changing rate of the energy dE/dt ?
- (c) What is the power input by the machinery calculated from $P = Fv$? Show that it is the input power that causes the energy to change, $P = dE/dt$.